

Q1. (1 points) Given a transformer with a model hidden dimension of 768 with 12 attention heads, what is the head hidden dimension?

The head hidden dimension is $\frac{768}{12} = 64$.

Q2. (1 points) Assume that we have a sequence length of s tokens, what is the per-token time-complexity of self-attention using the KV-cache?

The per-token time complexity of self-attention with KV-cache enabled is $O(sd)$ where d is the head dimension.

Q3. (2 points) During the training of Deepseek V3, what data-type are the optimizer master weights stored in?

"The master weights (stored by the optimizer) are still retained in FP32 to ensure numerical stability throughout training."

Q4. (2 points) Given a transformer with $L = 61$ layers, $h = 128$ heads per layer, and $d_k = 128$ head dimension, what is the total number of kv cache entries per token when using MHA and MQA.

For MHA, the number of entries generated is $128 \times 128 \times 61 = 999,424$ K-V pairs.

For MQA, the number of entries generated is $128 \times 61 = 7,808$ K-V pairs.

Q5. (4 points) Given a standard MHA architecture with $L = 64$ layers, $d_{model} = 8192$ model hidden dimension, $h = 32$ heads per layer, $d_k = 256$ head dimension, and $n_{tokens} = 65536$, what is the total number of parameters and memory required to store the weights in bf16, assuming an MLP up-down projection factor of 4? Exclude biases, layernorm, and positional encoding parameters. Focus primarily on the token embeddings, attention, and MLP modules.

	Number of Params	Memory Required
Token Embeddings	$d_{model} \times n_{tokens} = 8192 \times 65536 = 2^{29}$	$2^{29} \times 2 = 2^{30} \approx 1.07 \times 10^9$ bytes
Attention	$L \times (3 \times d_{model} \times (h \times d_k) + (h \times d_k) \times d_{model}) = 64 \times 4 \times 8192 \times 8192 = 2^{34}$	$2^{34} \times 2 = 2^{35} \approx 3.44 \times 10^{10}$ bytes
MLP modules	$L \times (2 \times d_{model} \times 4d_{model}) = 64 \times 2 \times 8192 \times 4 \times 8192 = 2^{35}$	$2^{35} \times 2 = 2^{36} \approx 6.87 \times 10^{10}$ bytes
Total	$2^{29} + 2^{34} + 2^{35}$	$2^{30} + 2^{35} + 2^{36} \approx 1.04 \times 10^{11}$ bytes or 97 GiB